



SEMESTER 1 EXAMINATIONS 2023/2024

MODULE: CA119 - Data Science and Databases

PROGRAMME(S):

DS BSc in Data Science
ECSAO Study Abroad (Engineering & Computing)

YEAR OF STUDY: 1,0

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 (Entity-Relationship Modelling)

[TOTAL MARKS: 25]

You will design a database for Dublin airport to capture analytics on flights, passenger numbers and destinations. The 5 entities are the **Airline**, **Flight**, **Destination**, **Passenger** and **BoardingCard**.

You may use the same E-R diagram to answer all questions but if you do, you must state explicitly which part of that diagram answers each of the parts a) to e) below.

Q 1(a) [5 Marks]
Consider between 3 and 5 simple attributes for each entity. Draw this version of your E-R diagram.

Q 1(b) [8 Marks]
Nominate one of the 5 entities as a weak entity. Explain:
I. Why it is a weak entity.
II. Which entity is its owner entity.
III. What type of participation constraint is used.
IV. What is used as the partial key.
V. What is the full key for this entity.

Q 1(c) [3 Marks]
Proceed by adding non-simple attributes as follows: 2 of the entities should have complex attributes and 1 entity should have a multi-valued attribute.

Q 1(d) [3 Marks]
1 entity should have a derived attribute. Explain why you chose that particular derived attribute.

Q 1(e) [6 Marks]
Add the remaining relationships. Explain why you gave each relationship its cardinality.

[End of Question 1]

QUESTION 2 (SQL Programming)

[TOTAL MARKS: 25]

This question uses the schema in Appendix A

Q 2(a)

[10 Marks]

Using the Customer Purchases table:

- (i) How many items were sold on 3rd July 2019 (2019-07-03)?
- (ii) What is the total cost of all products sold that day?
- (iii) What customers (customer IDs) were the top 5 spenders that day?

Q 2(b)

[7 Marks]

- (i) Write a query that INNER JOINs the vendor table to the vendor_booth_assignments table and then sort the result by vendor_name, and then market_date.
- (ii) Explain the semantics of that query in terms of what the result set contains.

Q 2(c)

[8 Marks]

- (i) Get the customer first and last names, the vendor name, product_id, and quantity for all purchases.
- (ii) Return the same 5 fields but this time, sum the quantity for each customer, for each product.
- (iii) Finally, filter these results to include only the 'Field of Corn' vendor.

[End of Question 2]

QUESTION 3 (Functional Dependencies)

[TOTAL MARKS: 25]

This question uses Appendix B.

Q 3(a)

[6 Marks]

Describe the method for determining all Functional Dependencies (FDs) with 2 attributes (**flight_num**,a) on the left hand side and 1 attribute (b) on the right side. In your description, write down all combinations of FDs in the order in which you find them. Note: a and b represent any 2 attributes $\neq$ flight_num.

Q 3(b)

[5 Marks]

Label each FD found in part (a) as Full or Partial.

Q 3(c)

[8 Marks]

Identify all functional dependencies with left hand sides:

- (i) loyalty_num $\rightarrow$ X (where X is a single attribute)
- (ii) passenger_id $\rightarrow$ X (where X is a single attribute)
- (iii) Are all of the rules in parts (i) and (ii) commutative (the reverse FD holds)? Explain your answer.

- Q 3(d) [6 Marks]
- (i) In terms of decomposition, what is meant by the lossless join property. Using the table in Appendix B, provide an example.
 - (ii) What is meant by the dependency preservation property? Using the same table, provide an example.

[End of Question 3]

QUESTION 4 (Schema Normalisation)

[TOTAL MARKS: 25]

This question uses the table in Appendix B.

- Q 4(a) [3 Marks]
- State the condition by which a table is in first normal form (1NF).
Is the passenger_flight table in 1NF? Explain your answer.

- Q 4(b) [6 Marks]
- State the condition by which a table is in second normal form (2NF).
Explain why the passenger_flight table is not in 2NF by providing 2 functional dependencies that prove that 2NF does not hold.

- Q 4(c) [5 Marks]
- Convert the passenger_flight table to 2NF and show the structure of the tables.

- Q 4(d) [6 Marks]
- State the condition by which a table is in third normal form (3NF).
Check that tables from part 4(c) are in 3NF.
Write an explanation for each table.

- Q 4(e) [5 Marks]
- Does the schema in 4c) require further decomposition?
If not, explain why.
If yes, explain the process of further decomposition.

[End of Question 4]

QUESTION 5 (Temporal Analysis)

[TOTAL MARKS: 25]

This question uses the schema in Appendix A

You require a dataset to analyse customer preferences:

- (i) Frequent customers ... with a top- N listing [Dataset C]
- (ii) Favourite vendors ... by the number of customers they get [Dataset V]
- (iii) How weather affects sales at the market [Dataset W]

Q 5(a)

[6 Marks]

For datasets C and V above which require aggregations for each product and vendor respectively, outline the individual steps required to build the datasets.

In your answer, explain why you selected the tables you did and describe the order of processing.

Q 5(b)

[6 Marks]

Use dataset C to determine customer loyalty by getting the difference between the first and last purchase dates for customers. Describe this MIN-MAX analysis:

- (i) How you would create this dataset?
- (ii) How can it be used to answer the question of how long a customer has been coming to the market?

Q 5(c)

[8 Marks]

Answer the following questions using SQL expressions together with a short explanation of why you composed the SQL expression.

- (i) On how many market days did it rain?
- (ii) Which are the most popular products on rainy days?

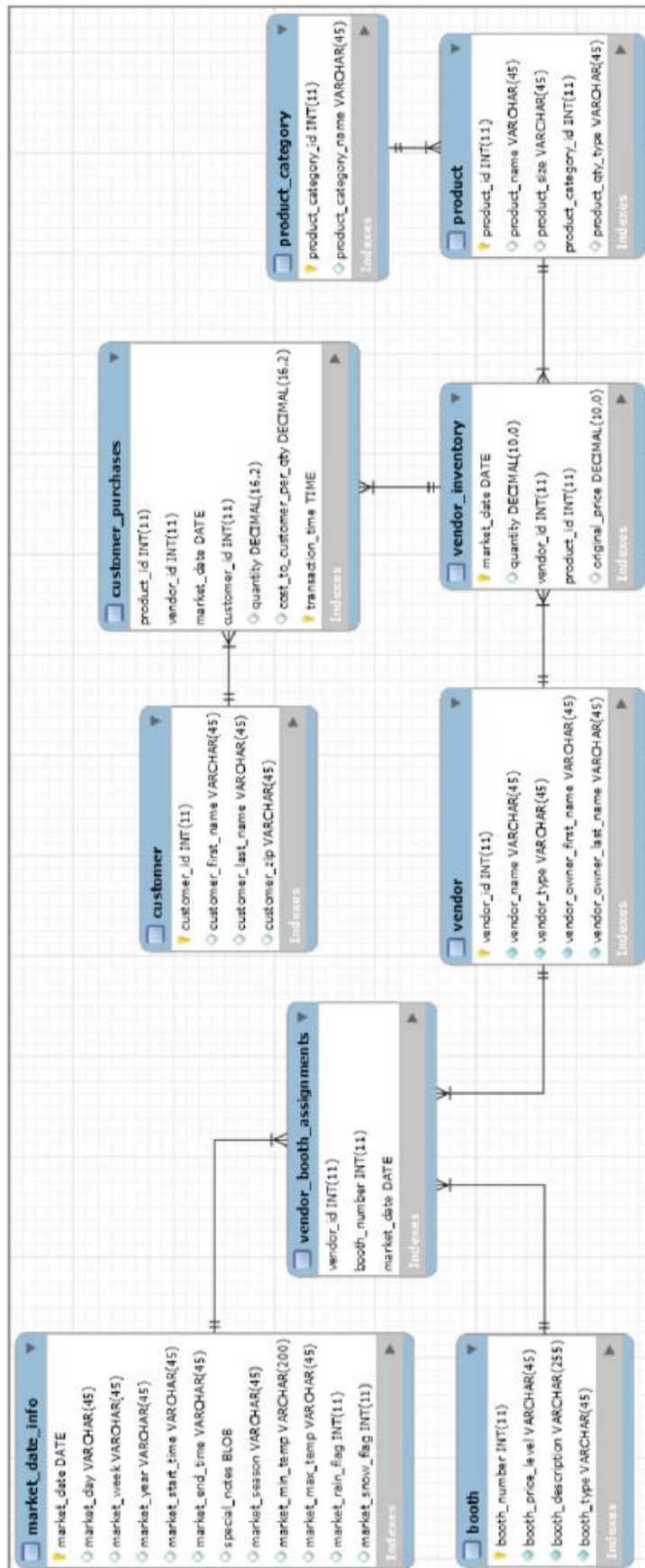
Q 5(d)

[5 Marks]

Explain how the RANK function operates. Provide an example from one of your datasets to show how RANK works.

[End of Question 5]

Appendix A: Farmer's Market Schema



Appendix B: passenger_flight

flight_num	passenger_id	date	destination	duration	loyalty_num
FR123	P101	01/02/2023	Bristol	1	2101
EI246	P102	01/02/2023	Alicante	2	2102
EI248	P103	01/02/2023	Amsterdam	1.5	2103
FR223	P104	01/02/2023	Dusseldorf	2	2104
FR153	P105	01/02/2023	Palma	2	2105
US401	P106	01/02/2023	Atlanta	7	2106
US402	P107	01/02/2023	Boston	6	2107
EI514	P108	01/02/2023	Toulouse	2	2108
EI646	P109	01/02/2023	Madrid	2	2109
FR129	P110	01/02/2023	Rome	2	2110
EI101	P111	02/02/2023	New York	7	2111
EI514	P101	02/02/2023	Toulouse	2	2101
FR333	P105	02/02/2023	Madrid	2	2105
EI249	P112	02/02/2023	Rome	2	2112
EI101	P120	02/02/2023	New York	7	2120
US401	P121	02/02/2023	Atlanta	7	2121
EI102	P122	02/02/2023	Boston	6	2122
IB400	P102	02/02/2023	Alicante	2	2102
FR102	P123	02/02/2023	Amsterdam	1.5	2123
FR223	P108	02/02/2023	Dusseldorf	2	2108

No passenger will depart Dublin airport twice on the same day!

[END OF EXAM]